Delivery 1: Dashboard

General information

In groups of 2 or 3 people prepare a dashboard with two screens: one main screen with global information, and one additional screen with detailed information on a particular aspect of the dataset. The graphic may perfectly be static, but if desired, you may include interaction.

This delivery accounts for 40% of the final qualification. It should be submitted in time. The submission should only consist of one zipped file on the corresponding task on Campus Virtual with the necessary content for the task with the name DashboardTaskMember1Member2Member3Member4.zip, only one member of the group has to send the task.

As a basis, you can use the provided datasets or any other previously negotiated with the professor. You shall mention the source of the dataset and any re-used code that has served as inspiration.

Educational Goal

In this task you are expected to put into practice best practices, good chart selection decisions and perception principles. The process and decisions are as important as the final implementation, so pay attention to them.

You may plan a more ambitious dashboard and then implement a simplified version, to equilibrate capacity on design and skills to implement.

You may use AI to build the dashboard as long as you clearly acknowledge where and how did you use it and are capable of answering any questions about the code or the decisions. Deliveries using AI are subject to an interview with all members of the group.

Tasks

For each part, as an orientation some questions are provided that should be answered by the given information; but the delivery is not expected to be a question-answer text, nor a list of bullets.

Part 1: Exploratory Data Analysis (EDA): Data (bottom-up): Study and describe the data + explore relationships 15%

a) the data

This part shall describe the data in a thorough way in order to plan the visualization from a bottom-up perspective. Analyze the data and the associated documentation. For each field, clarify what type of data it is, whether it has a temporal or geographic nature, what its value range and domain are, which units it uses, and what level of precision is required. Examine the distribution and identify outliers. What is the currency or timeliness of the data? How often should it be updated? Identify groupings and relationships among the different fields and their

characteristics. Create several charts to explore distributions and relationships. Look for missing data, incoherences, real or forecast data… Comment on the findings.
Based on this exploratory study, determine which fields to include in the final visualization and which relationships to highlight. Deliver a detailed description of the data and the exploratory charts.

Part 2: Audience (top-bottom): Who is your audience? 15%

This part shall describe the audience in a thorough way in order to plan the visualization from a top-down approach. Are they experts in the use of charts? Which platform (mobile, PC, tablet – choose only one --) will they use to see your charts? [I recommend tablet or desktop computer for this initial task] How often will they use your dashboard? Describe a persona, their goal and a scenario of use of your visualization

Based on the dataset and the audience, describe the goals of the dashboard and state 5 to 7 questions your dashboard must answer.

The questions shall be written from the perspective and needs of users.

Part 3: Reflection and adjustment loop 5%

Review your decisions and needs and consider if you need additional data, if you need to merge the available data… or other adjustments. Briefly describe any small changes you do to Part 1. If you incorporate new data, describe it here in detail.

Part 4: Selection of chart and encoding 15%

Decide the type of charts and the encoding of the data taking into account the Dashboard's goals, the function of the charts and the available data. Describe decisions on encoding and selection of charts, layout, based on perception properties and Gestalt laws, visualization principles and best practices. Take into account accessibility issues.

Create a prototype design, low-fi, and explain how this design will answer the proposed questions. Argument decisions.

Part 5: Interaction planning 15%

Complete the prototype with interaction. Describe any interaction within each chart or between charts. Describe the interaction mentioning the particular and big theoretical model seen during the course.

Explain decisions of interaction.

Part 6. Implementation and delivery 30%

Prepare the final dashboard on an html page with highcharts, jupyter notebook with vega-altair, or create a PowerBI or Tableau dashboard.

Create a hi-fi visualization, explain implementation decisions.

Deliver explanations and samples for all the process, the final presentation to be seen locally, the notebooks and the datasets.

You may deliver an advanced draft of your dashboard 15 days before the final delivery to opt in for a virtual tutoring session and receive feedback.

Part 7. Work group and personal reflections – 5%

Briefly comment how you have organized the tasks within the group, how was the process of creating the visualization, if the theoretical models did help in creating the visualization and any surprises or difficulties of the task.